

Table 1: **Results on RMBench [1].** We report the success rates over 100 rollouts.

Task	$\pi_{0.5}$	FastWAM	Lingbot-VA	Ours
Observe and Pick Up	9%	0%	13%	27%
Rearrange Blocks	13%	0%	100%	100%
Put Back Block	11%	0%	100%	100%
Swap Blocks	24%	0%	99%	100%
Swap T	15%	7%	88%	94%
Battery Try	16%	20%	41%	41%
Blocks Ranking Try	6%	26%	100%	100%
Cover Blocks	0%	0%	79%	98%
Press Button	0%	0%	84%	87%
<i>Average</i>	10.4%	5.9%	78.2%	83.0%

importance of long-term memory. MemoryWAM further improves the average success rate by 4.8 percentage points over LingBot-VA and achieves leading performance on every task. This suggests that retaining all historical tokens is not the only effective way to support persistent memory: by preserving full tokens for key observations and compressing long-range history into gist tokens, MemoryWAM retains task-relevant context in a more compact form.

4.4 Real-World Experiments

Our hardware platform consists of an ARX dual-arm robot and a RealSense D455 camera that provides RGB observations. We compare MemoryWAM with two representative baselines: $\pi_{0.5}$ [62] and LingBot-VA [7]. We design two challenging memory-dependent tasks, *Shell Game* and *Look and Press*, as shown in Fig. 5. In *Shell Game*, the robot should identify the cup covering a small cube after a human swaps the cups. In *Look and Press*, the robot observes two numbers on the table, presses the left and right buttons the corresponding number of times according to the observed numbers, and finally presses the rear button once to indicate completion.

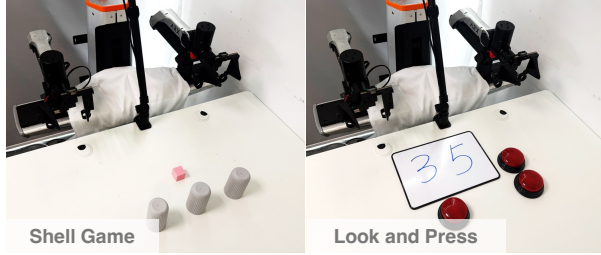


Figure 5: Illustration of the real-world tasks.

The results are reported in Tab. 2. Consistent with the simulation experiments, policies with only a short observation window struggle on memory-dependent tasks, while LingBot-VA improves by retaining full history. MemoryWAM achieves the best performance on both tasks with substantially lower latency and GPU memory cost than LingBot-VA. Notably, the high inference latency of LingBot-VA causes it to miss the cup swaps in the *Shell Game*, leading to task failure. These results demonstrate that the proposed hybrid memory not only improves efficiency but also provides a more compact and task-relevant memory representation for real-time, long-horizon robotic manipulation.

Table 2: **Results of real-world experiments.** We report the number of successes over the total number of trials.

Task	$\pi_{0.5}$	Lingbot-VA	Ours
Shell Game	5/20	13/20	18/20
Look and Press	0/20	14/20	15/20

4.5 Effectiveness of Design Choices

To systematically validate the necessity of each component in the proposed hybrid memory, we conduct ablation studies on two challenging tasks from RMBench [1], *Cover Blocks* and *Press Button*. We compare MemoryWAM with four variants: (1) *w/o Anchor Frames*, which removes the original video latents corresponding to event-boundary observations from the context and uses gist tokens as a substitute; (2) *w/o Gist Tokens*, which removes the long-term gist memory; (3) *w/o Sliding Window*, which removes the original video latents corresponding to recent frames from the context and uses